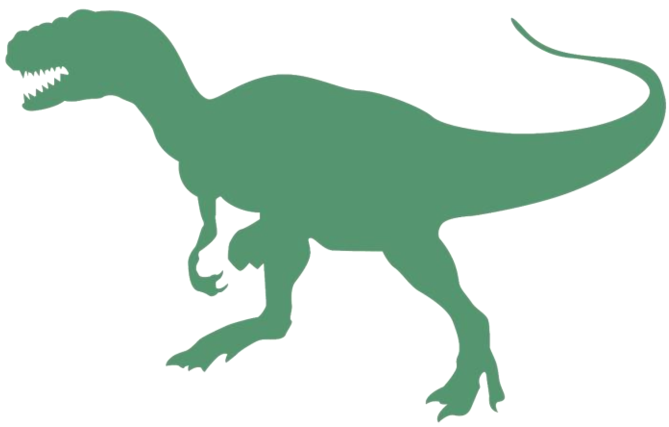
Universität Bern, CAS Natural Language Processing - AI for Language

t-RAG: Terminology Precision in Multilingual RAG



A Case Study on Swiss Railway Operating Regulations

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Abstract

Retrieval-Augmented Generation (RAG) is increasingly used to ground LLM outputs in domain-specific knowledge, but much research and tooling have focused on monolingual, English-centric settings. Results cannot necessarily be transferred to multilingual settings which are often very different, depending on the languages, documents and use cases involved. For organisations operating in high-stakes domains, such as the Swiss Federal Railways (SBB), precise and consistent terminology across German, French and Italian is required. This project investigates whether a multilingual RAG setup (indexing documents in all three languages) produces more precise and consistent terminology than a monolingual (German-only) setup, applied to the Swiss railway operating regulations (FDV/PCT). Two RAG pipelines combining semantic search, BM25 lexical scoring and cross-encoder reranking were built and compared on two use cases: classic content Q&A in three languages and cross-lingual terminology queries, the latter used as an exploratory stress test of the pipelines' limits. The use cases were evaluated for retrieval accuracy, terminology precision and overall answer quality.

Retrieval performance proved to be the decisive factor: the multilingual setup achieves a hit rate of 95.5% for French and Italian queries, against 36% and 54.5% for the monolingual setup, and terminology precision in the generated answers follows directly from this. When the correct source document is unavailable in the query language, the generative model fails to produce the correct domain-specific term, confirming that an LLM's parametric knowledge cannot always substitute for in-language grounding. The cross-lingual terminology queries expose a more fundamental limitation: because these queries interrogate the linguistic structure of the corpus rather than its factual content, the RAG retrieval paradigm itself is mismatched to the task. The findings suggest that multilingual RAG is a viable tool for content-oriented Q&A in regulated multilingual environments, but that terminology lookup for language professionals requires a different architecture, such as more recent approaches like Terminology-Augmented Generation (TAG) that integrate dedicated terminology resources dynamically.

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Abbreviations

SBB	Swiss Federal Railways (Schweizerische Bundesbahnen)
FOT	Federal Office of Transport
FDV	Fahrdienstvorschriften, Swiss railway operation regulations
PCT	Prescriptions de circulation des trains (fr), Prescrizioni sulla circolazione dei treni (it), Swiss railway operation regulations
RAG	Retrieval-Augmented Generation
TAG	Terminology-Augmented Generation
LAG	Lexical-Augmented Generation
LLM	Large Language Model
TTR	Type-Token Ratio
ChromaDB	Open source embedding database using HNSW indexing.
HNSW	Hierarchical Navigable Small World, an algorithm for approximate nearest neighbour search.
ANN	Approximate Nearest Neighbour
CAT	Computer-Assisted Translation
MT	Machine Translation

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1. Introduction

Large Language Models (LLM) are powerful tools for text generation, they produce extremely fluent outputs but their intrinsic knowledge is limited to the traces that the training data left in their parameters. Over the last few years, Retrieval-Augmented Generation (RAG) has been established as an efficient technique to alleviate some of the basic problems caused by the LLMs' structural limits, including hallucinations or the lack of domain-specific knowledge. It consists of providing external, up-to-date information to the parametric knowledge of an LLM: for every query, a specific context is constructed with input from a database. The LLM thus uses the retrieved information together with the query itself to generate an appropriate answer.[1]

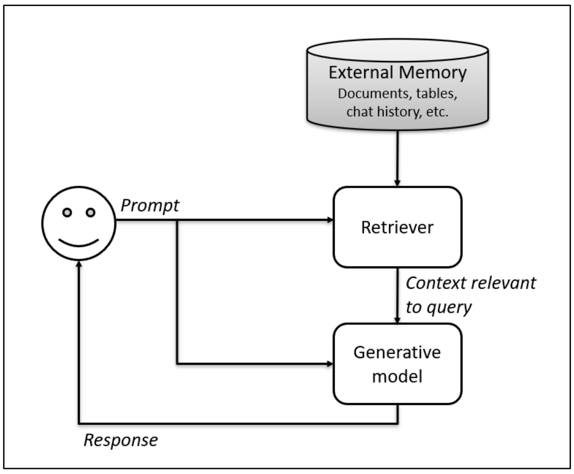

Figure 1: Basic RAG archtecture (based on Huyen 2023, p.256).

While the technique is widely used today, many of the previous efforts have focused on monolingual and specifically English language contexts and their findings cannot necessarily be applied directly to multilingual situations, depending on the language(s) present in the knowledge database and the language(s) used for querying.[2] Multilingual RAG faces challenges in retrieval because of language variation between queries and documents, sometimes there are inconsistencies when different language sources contradict each other.[3] Language preference can also be an issue, English being the dominant language across most of the training data used.[4] Each multilingual context is different and consequently the solutions need to be adapted to each situation's specific requirements.

In the multilingual context of Switzerland and more specifically the Swiss Federal Railways (SBB), English is not the dominant language.[5] The company languages are German, French and Italian, and the SBB Language Service is the competent unit for the multilingual internal and external communication.[6] Its main responsibility is maintaining quality translation and high-performance multilingual resources such as an extensive terminology database and specifically trained engines for Machine Translation (MT). These resources are not just nice to have but extremely important, even more so as many of the texts treated in the SBB Language Service have a direct impact on the company image (e.g. marketing campaigns) or are security and even safety relevant (e.g. operation regulations). With the advent of generative AI, new technologies such as LLMs and RAG have come into focus but even though these technologies may solve many problems, they are not optimised for terminological precision or multilingual consistency, something that is essential when working as a language specialist in high-stakes domains such as railway operations.[7] This raises the central question this project addresses: does a multilingual RAG setup produce more precise and consistent terminology than a monolingual one, and where does the architecture break down structurally when pushed across language boundaries?

This research project, by building multilingual terminology RAG setups, pursues two complementary goals applied to Swiss railway operating regulations (FDV): The first is to compare a monolingual (German-only) vs. a

[1] Lewis et al. 2021, p.1 and Huyen 2023, p.255.

[2] Lewis et al. is a good example for the initially English-centric research in this field.

[3] See Chirkova et al. 2024, Park & Lee 2025 or Ranaldi et al. 2025.

[4] Wu et al. 2024.

[5] The Swiss national and official languages are defined in the Federal Constitution of the Swiss Confederation (https://www.fedlex.admin.ch/eli/cc/1999/404/en) and the Federal Act on the National Languages and Understanding between the Linguistic Communities (https://www.fedlex.admin.ch/eli/cc/2009/821/en). NB: Translation in Switzerland is also a politically sensitive issue. The majority of the population is German-speaking, many publications are therefore written in that language and subsequently translated. The French and Italian-speaking communities often feel excluded or discriminated against when the translations are substandard or even non-existent. This reality is also present within the SBB.

[6] The Romansh language is not an official company language because the SBB does not run trains in that linguistic area of the country.

[7] Di Nunzio 2025, 97.

multilingual (de/fr/it) RAG setup on operational content queries, establishing which architecture better serves users asking factual questions in their native language. It represents a use case of classic Q&A, e.g. for better understanding of the concepts or procedures in the documents. We expect that the answer quality and terminology precision of the multilingual setup is better for French and Italian queries because the answers are grounded in documents of the same language featuring the correct terminology. In the monolingual setup, the LLM would generate a French or Italian answer based on a German document, the translation task would be done by the generative model without target language grounding.
The second use case represents a stress test for both pipelines, using cross-lingual terminology queries to probe where it succeeds and where adaptations may be necessary. This second part is explicitly exploratory: a conversational RAG interface would not be a translator's first tool for single-term lookup, dedicated terminology databases integrated directly in their work environments such as CAT tools serve that need. As we will see, cross-lingual queries expose pipeline behaviour and challenges that content Q&A alone cannot reveal, and the test result clearly shows the limits of such tools. As it is, a terminology RAG might not be the most suited tool for these use cases after all.[8]

The present paper details the different RAG setups and evaluations conducted to pursue these research questions. It presents the different steps and results of the project in the following sections: Chapter 2 presents the data used for the RAG databases as well as the different evaluation test sets. Chapter 3 presents the implementation of the two RAG setups, including the different components and pipelines for retrieval and generation. Chapter 4 presents the evaluation design and methodology and in chapter 5 we will look at and discuss the results. Finally, in chapter 6 we will draw possible conclusions from the project results. Examples from the test data and the generated results as well as the project structure as available on GitHub can be found in the Annexes.

2. Data

This project has been realised using the Swiss railway operation regulations, published by the Federal Office of Transport (FOT). The regulations contain safety-related processes for all types of rail operations and apply to all railway companies operating on Swiss railway infrastructure. They are generally further clarified through each company’s internal implementing provisions. We are thus working within the domain of prescriptive regulatory text with tightly controlled, officially defined terminology. The regulation documents are publicly available in German (Fahrdienstvorschriften, FDV), French (Prescriptions de circulation des trains, PCT) and Italian (Prescrizioni sulla circolazione dei treni, PCT) on the FOT website. In Switzerland, most federal laws and regulations are co-written in French and German but also translated into Italian (and Romanche where necessary). The three documents are therefore all legally binding, the different language versions are high quality human translations by subject matter experts rather than MT and are considered equivalent.
This data was chosen because it is easily accessible, available in several languages and high quality, but the documents contain technical, domain-specific content with precise terminology rather than general language content. Driving trains is an extremely technical and safety-relevant job, for passenger as well as cargo trains and shunting operations. Communication and execution errors can lead to serious financial, reputational and human consequences. It is therefore crucial that all railway companies in Switzerland understand and use the same procedures and language. More interestingly, the content also has to be understood and worked with, by operational staff but also by language specialists, justifying practical applications and use cases such as RAG (e.g. translators who need to understand the content in order to translate it, or future train drivers tracking their learning progress for upcoming exams).

[8] The name t-RAG also evokes the mighty T-Rex, a powerful animal in its domain but with notably short arms - just a little too short when it comes to reaching across (language) barriers.

Corpus data

The operation regulations are composed of 15 chapters, each with their own regulation number (R 300.1 through R 300.15). The German document is the shortest, Italian is longer by around 7.5% and French by more than 11%, which reflects the general structure of these languages. This tendency is also visible in the chapter length, which varies between ~4000 and more than 100 000 characters (see Figure 2). Some of them contain tables, calculations, figures or form templates.

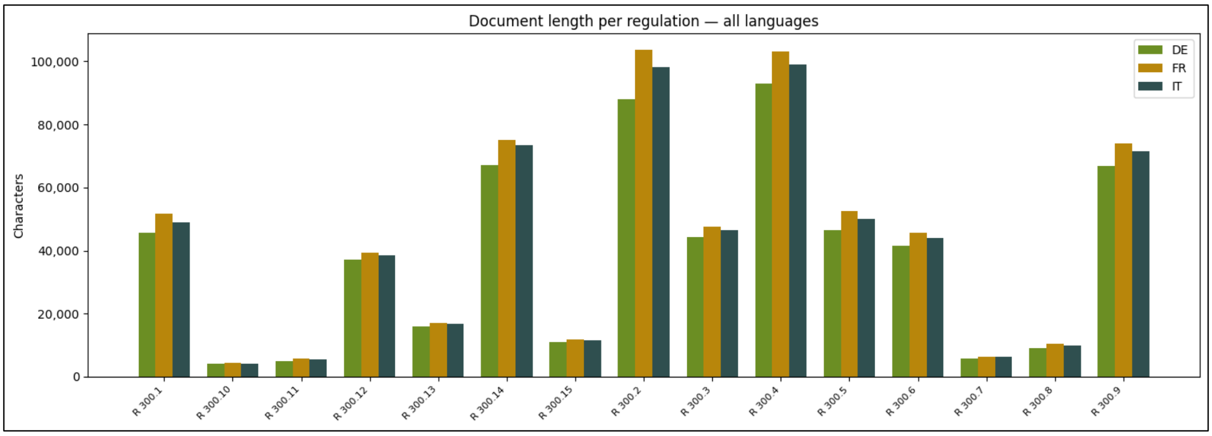

Figure 2: Document length per regulation for all languages.

Further statistics like sentence length also confirm these language characteristics, while the number of unique content words points to another particular aspect of the German language, compound words. French and Italian use less than 5000 unique words, 4646 and 4918 respectively, the German text however contains more than 6783 unique words. Compounding, amongst other characteristics of this language, tends to create more unique words, a fact that has to be considered for text preprocessing and chunking rules.

The **vocabulary** clearly shows the technical railway background of the texts - the most frequent unique content words being concepts like signal, train, vehicle or brake. It also shows that the lexical overlap (cross-lingual mapping) is very small across languages, the only term amongst the top 20 to appear in more than one of the three languages being *"signal"*, which is spelled identically in German and French. The **TF-IDF signature** also confirms that each of the 15 regulations is anchored around a compact, distinctive set of terms that cleanly identifies its subject matter. This reflects the structure and characteristics of prescriptive regulation, rather than general language, each chapter defining and applying a specific subset of the controlled terminology, with little topic drift.

The **terminology density** analysis also shows similar characteristics. Type-Token Ratios are moderate across all three languages, reflecting the fact that authors and translators are legally bound to use the officially defined term for a concept every time it appears. The mean **Gini coefficient** values also confirm moderate to high concentration - a small number of terms account for a disproportionate share of usage in each regulation. Figure 3 shows that the highest values can be found in very technical regulations about railway signals (R300.2), brakes (R 300.14) or shunting movements (R 300.11).

Language	Characters	Words	TTR	Gini
DE	580'805	74'753	0.39	0.43
FR	647'942	101'856	0.3	0.54
IT	624'724	94'902	0.32	0.53

Table 1: Corpus statistics for all languages.

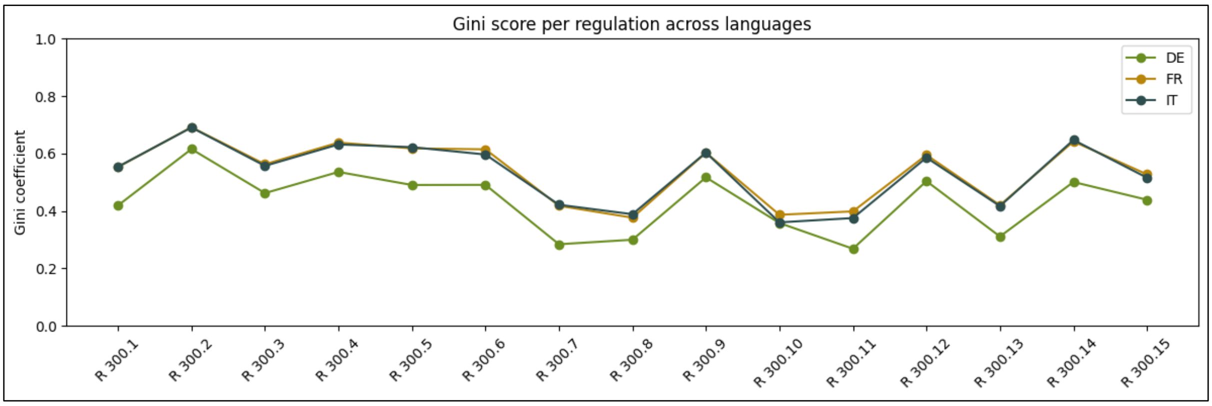

Figure 3: Gini score per regulation for all languages.

Both TTR and Gini reflect a vocabulary that is simultaneously large (many distinct technical terms) and internally repetitive. Unlike journalistic or literary texts, where writers avoid repetition through paraphrasing, regulatory authors are legally bound to reuse the same term every time. This consistency should generally be favourable for both lexical and semantic retrieval. Both mechanisms will find a strong, consistent signal in each chunk and the risk of a relevant passage being retrieved weakly because its key terms happen to be sparse is low. Where the Gini score is particularly high, retrieval precision should be especially reliable since the dominant terms are highly discriminative. However, as Soman & Roychowdhury (2024) have found, in technical documents other factors such as chunk size, acronyms or even the structure of the documents also have an influence on retrieval success, and even high similarity scores can be misleading in long documents, when many chunks with almost identical terminology are found. The content of the documents in itself is therefore no guarantee for good retrieval performance.

The documents are available in PDF and MS Word **format** and their **structure** is good overall. This is also because they are published regulatory documents and therefore contain hardly any typos, noise (e.g. code, markup) or otherwise problematic content. Some preprocessing steps were necessary nevertheless before the files could be used for indexing: The Word files (.doc) are formatted as tables and contain leftovers from macros and other special templates used for regulatory texts in the Federal Administration. They were therefore manually converted to plain text (.txt) for this project. Some of the tables and form templates in the sections were removed as they would not have been readable for the model. In one of the chapters in Italian, a section was missing and it was added manually to ensure consistency.

The documents feature a consistent and parallel document structure featuring tab-delimited section headings (hierarchical section structure and section IDs across all three languages). However, some of the files presented inconsistent formatting of these section headings, probably caused by the conversion of the table formatting. These section headings were also corrected for consistent chunking, i.e. to match the regex pattern for section heading detection (see Document processing below).

Evaluation data

In order to evaluate the different components and RAG setups, two separate annotated test sets were curated manually, based on the document contents. First, a **list of potential terms** was identified, with a focus on terminology variation. Terms were chosen e.g. if they are used in a particular way specific to the geographic region or the railway operation domain, or if confusion with a similar term is possible. For example, a *"segnale basso"* (it, Switzerland) could be called a *"marmotta"* in Italy, or even in the Italian speaking part of Switzerland a *"segnale nano"* (informal analogy to the German *"Zwergsignal"* and the French *"signal nain"*). Another good example would be the concepts *"aiguille d'entrée"* and *"première aiguille"* (fr) that are semantically close but should be clearly differentiated. Given the specificity of the domain and the intra-linguistic variation, it is highly unlikely that a generative model would be able to use the correct terminology consistently when only using its parametric knowledge. At the same time, given the high-stakes domain, it is important that a useful RAG system

would generate answers featuring the correct terms. In those setups where query and document languages do not match, these terms will therefore be a clear indicator of how good these systems' terminology precision is.
The final list of 24 terms was reviewed by native language specialists of the SBB Language Service before being used to create the test sets,[9] though not all of them were used in the queries.

The **first test set** contains 22 questions, each in three languages, to make a total of 66 queries, structured as same-concept triplets (de/fr/it versions of the same query). Each query is annotated with the expected target regulation and section and target terminology and was reviewed by native language specialists of the SBB Language Service. The queries represent standard content questions such as *"Welche Fahrzeuge dürfen nicht in Züge eingereiht werden?"* (de) or *"Dove si trova la soglia di velocità in stazione se sul lato d'entrata la velocità in stazione è diversa dalla velocità di tratta?"* (it).[10] The identified target terms are not themselves present in the query, to avoid explicitly providing the correct terminology in the prompt. They should however be present in the generated answer, provided the retrieval mechanism finds the expected target regulation and section.[11]

The **second test set** contains 18 queries in different language combinations, again based on the identified terminology. They were also annotated with the source and target language, target terminology and expected target regulation and section. Here, the focus is on cross-lingual queries, in order to test the direct retrieval of terminology. The queries include questions such as *"Wie lautet der offizielle italienische Fachbegriff für Abstossen?"* (de -> it) or *"Comment appelle-t-on les signaux nains en allemand?"* (fr -> de). [12] The source language being the language of the query, the target language that of the term to be found.

3. Implementation

This chapter will present the different components of the monolingual and multilingual RAG setups used in this project. Figure 4 shows that the architectures and use cases are designed identically. Strictly speaking, the monolingual setup (mono_rag) still uses multilingual embedding and a multilingual generative model, the distinction from the multilingual setup (multi_rag) lies only in the contents of the document database which is in German only vs. all three languages (de/fr/it).

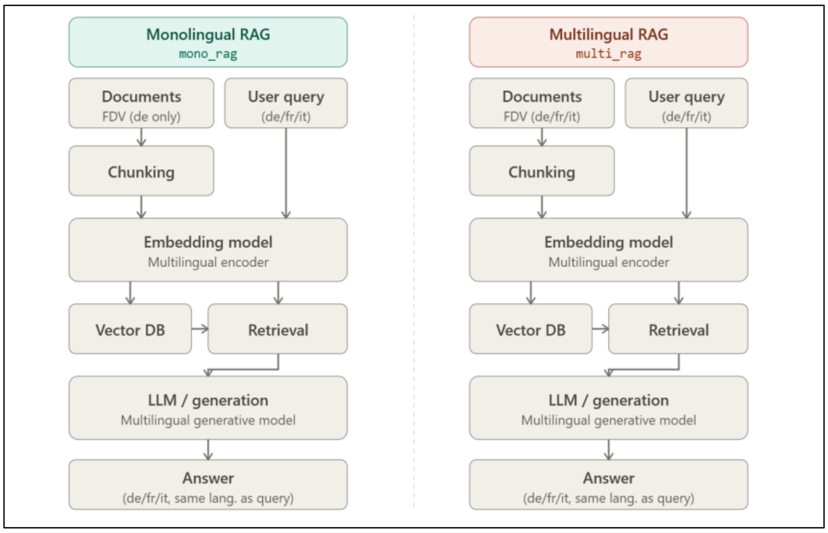

Figure 4: RAG architectures for monolingual and multilingual setup.

[9] For the full list see Annex 1.
[10] For more examples see Annex 2 or project file *questions.csv*.
[11] Queries were generally formulated with reference to titles or very close to the wording in the original document so as to facilitate retrieval. The goal of this use case is to explore terminology handling, not to explore edge cases with explicitly difficult phrasing, and generalised retrieval failure would make further analyses impossible.
[12] For more examples see Annex 2 or project file *cross_lingual_questions.csv*.

We will now look at the RAG components and the choices made for this specific project.

Document processing

The documents being available in raw text format and generally high quality, only minimal preprocessing was necessary for text normalisation: UTF-8 BOM removal and handling of soft-hyphen and non-breaking spaces, as these could interfere with a potential lexical search (e.g. in German compound words like *“Schutz***\xad***strecke”*). A three-pass chunking process was implemented that would prepare the text for the embedding process:

1. Split on section headings
2. Merge stubs smaller than 150 characters (`CHUNK_MIN_CHARS = 150`)
3. Re-split oversized chunks larger than 800 characters with a 100-character overlap (`CHUNK_MAX_CHARS = 800` and `CHUNK_OVERLAP_CHARS = 100`). In very few cases (in French and Italian slightly more than in German), chunks would be longer than 800 characters, namely if a section is longer than the maximum length but the cutoff would then be merged back to the previous chunk because shorter than 150 characters.

The same function would save the following metadata for each chunk: `regulation_number`, `section_id`, `section_title`, `language` and `source_file` (including a repetition of the section header for follow-up chunks when a section is too long).

This chunking strategy means that the document structure is preserved and the chunks feature specific metadata for easy identification, thanks to the consistent formatting of the section headers and using a simple rule-based approach (regex) rather than splitting the text only based on character length. Splitting chunks at section headings or ends is also consistent with the regulatory characteristics of the documents, as has been shown above (see TF-IDF signature under Corpus data). However, this also means that the chunks vary in length from 150 to 800 characters (see Figure 5).

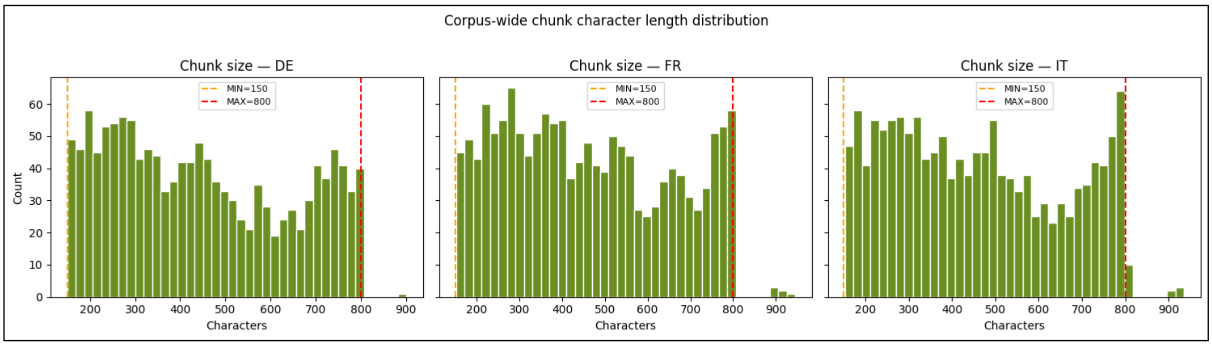

Figure 5: Chunk character length distribution for all languages.

Language	Characters	Words	Chunks
DE	580’805	74’753	1’360
FR	647’942	101’856	1’482
IT	624’724	94’902	1’430

Table 2: Corpus volume and chunks for all languages.

For the chunk sizing, several aspects were taken into consideration. The embedding model needs enough text to build a meaningful semantic vector. With anything below 150 characters (2-3 sentences of regulatory text) the embedding becomes too generic and retrieval will be unreliable. Generative models also have an input token limit and different languages do not necessarily have the same character/token ratio (around 4-5 characters per token for English). 800 characters per chunk remain safely within the limit. If the chunks were a lot longer, the text might be silently truncated in the embedding process.

At the same time, keeping chunks under 800 characters also allows the context to stay focused, and the LLM can attend to all of it. Very long chunks tend to dilute the relevant content with surrounding text, even more so when working with technical documents (see Corpus data above, or Soman & Roychowdhury 2024), which degrades answer quality even if the model technically has a large enough context window to fit it in. Table 2 shows the final number of chunks per language, again reflecting the overall size of the corpus per language.

Embedding and retrieval

One of the specificities of multilingual contexts that has to be considered when building RAGs is the potential linguistic discrepancy between documents and queries. A valid strategy might therefore be to translate documents into one or several languages before creating the vector embeddings, in order to reduce linguistic variation. However, introducing another translation step at this stage can also introduce errors and inaccuracies because the process relies heavily on the translation quality of the model employed.[13]

A different strategy consists of using a multilingual embedding model, where all queries and documents remain in their original language but are encoded and mapped to the same vector space.[14] The concepts of *"Sperrsignal"* (de), *"signal de barrage"* (fr) and *"segnale di sbarramento"* (it) would be embedded very close to each other, making cross-lingual retrieval possible.

Embedding

For this project the **multilingual embedding model** *intfloat/multilingual-e5-small* has been chosen, as it was specifically trained for Q&A and for asymmetric search, meaning that it is able to match different length passages, shorter queries to longer chunks.[15] It has an embedding vector size of 384, and a maximum input length of 512 tokens, large enough to treat the bigger chunks as well as query. It supports 100 languages, including the ones used in this project.

The vectors are stored and indexed in ChromaDB, a lightweight, persistent **vector database** that uses an HNSW (Hierarchical Navigable Small World) index. HNSW is a graph-based approximate nearest-neighbour (ANN) structure that organises vectors in a layered hierarchy: upper layers act as a coarse navigation map, while lower layers store the full neighbourhood graph. At query time the search enters at the top, greedily descends toward the query vector, and refines the result in the densest bottom layer.

The **collection** is configured with `hnsw:space: "cosine"`, so distances are measured by the angle between vectors rather than their absolute magnitude. This makes it more robust when treating different lengths of passages (e.g. queries and chunks), but it is also particularly appropriate in a multilingual corpus where the same concept may be expressed with very different surface forms across German, French, and Italian. By measuring directional alignment, the index surfaces semantically equivalent passages regardless of vocabulary or document length.

Retrieval

The retrieval pipeline operates in three successive passes, each narrowing down the candidate pool:

1. **Semantic search**: the query is encoded by the same multilingual E5 model used at indexing time and submitted to ChromaDB, returning the 30 most similar chunks (cosine distance via HNSW ANN search).
2. **Lexical search**: a BM25Okapi index, built over the same chunk corpus, scores all 30 candidates for lexical overlap with the query, after stripping punctuation from both query and text to avoid retrieval failures on exact matches.

The two scores are combined into a hybrid score, weighing 60% semantic vs. 40% lexical, and the top 6 candidates are retained.

3. **Reranking**: a multilingual cross-encoder (*mmarco-mMiniLMv2-L12-H384-v1*)[16] receives each of the 6 pairs (query and chunk) concatenated as a single input and produces a relevance score, i.e. the model scores which of the candidate chunks is the most relevant to the query.

The top 2 chunks by cross-encoder score are then retained for generation (`TOP_K = 2`).

[13] As noted e.g. by Park & Lee 2025, p.9.
[14] Jonietz & Zatta 2024.
[15] See https://huggingface.co/intfloat/multilingual-e5-small, Brezeanu 2024 and Alkestrup 2023..
[16] See https://huggingface.co/cross-encoder/mmarco-mMiniLMv2-L12-H384-v1.

The final pipeline was defined incrementally. Pure **semantic search** was the starting point, but initial testing revealed that for some queries the most pertinent passage was consistently absent from the top results. The underlying cause is probably a property of the corpus already visible in the Gini analysis (see Corpus data). Because the terminology is tightly controlled and concepts are distinguished by precise compound terms (such as *"Zugfahrstrasse"* vs. *"eingestellte Zugfahrstrasse"*), many chunks share very similar overall embedding distributions. The specific discriminating term (e.g. *"eingestellte"*) is diluted by the surrounding regulatory boilerplate, causing genuinely relevant chunks to cluster closely together in the embedding space and making the ranking among them unreliable.

Adding BM25 **lexical scoring** addressed this problem directly: an exact match on a controlled technical term produces a strong BM25 score regardless of how similar the surrounding content looks. Since there is little paraphrasing in the documents, this should prove to be a highly reliable signal.

Improvement on retrieval was immediate, especially pronounced for queries where the query language matched the document language in the database, i.e. German queries on mono_rag and all language queries on multi_rag. At the same time, this third pass had no negative impact on the other use cases (e.g. a French query on mono_rag with a BM25 score of 0).

The **cross-encoder** was added as a third pass to handle cases where neither embedding similarity nor term frequency is sufficient to discriminate clearly between candidates, for example when a query asks about a specific operational scenario and several chunks discuss related but distinct scenarios using near-identical vocabulary. The cross-encoder solves this by reading the query and each candidate chunk as a single concatenated sequence, allowing full bidirectional attention between query tokens and passage tokens.

However, even so the candidate pool had to be adjusted several times to find the optimal settings: with a pool of only 6 candidates initially, the most relevant chunk was sometimes eliminated by the hybrid score before the cross-encoder could evaluate it, precisely because there were too many candidates with high scores. The semantic retrieval pool was expanded to 30 candidates (`SEMANTIC_POOL = 30`) before applying BM25 and the rerank pool to 6 (`RERANK_POOL = 6`), which gave the cross-encoder sufficient coverage to consider relevant passages that ranked lower in the hybrid score.

Generation

The generation component runs entirely through Ollama, a local model serving framework, meaning no data is transmitted to external APIs. The model used for generation is *gemma3:4b* (3.3 GB), running locally through Ollama.[17] The number of chunks passed to the model is set to 2 (`TOP_K = 2`) and temperature is set to 0.1.

Running the system entirely locally using open models would make it fully reproducible without relying on third-party services. However, this choice also imposed constraints given the available local infrastructure. The **selection of the generative model** was thus mostly guided by hardware limitations and involved systematic comparison of several multilingual models through Ollama: *Mistral 7b* (4.4GB) produced strong outputs but required 6 to 7 minutes per query, making it impractical. At the other end, *Llama3.2 1b* (1.3GB) answered within 30 to 90 seconds but clearly showed insufficient instruction-following capacities for a grounded generation task. Several others were tested (e.g. *Phi4-mini*, *qwen2.5:3b* and *Aya23*)[18] and the final model choice struck the best balance across the three relevant dimensions in this project: answer quality, instruction compliance and response speed.

The comparison illustrates a trade-off that is particularly visible in RAG systems: retrieval pipeline improvements are necessary but not sufficient, a model that cannot reliably follow grounding instructions will still produce unreliable answers regardless of how well the context was retrieved. For a research setup, evaluation with a larger model is tractable because it runs offline in batch, while for interactive Q&A, response latency becomes a practical constraint. Reducing TOP_K to 2 was also a measure to reduce response time without having too much of an impact on the overall answer quality.

[17] See https://ollama.com/library/gemma3:4b and https://huggingface.co/google/gemma-3-4b-it.

[18] For model descriptions see https://ollama.com/search.

The **system prompt** (see Figure 6) is designed around the constraints of the domain rather than general conversational conventions. It was refined according to the results while testing and includes clear, explicit instructions necessary especially when working with smaller models. The instructions include grounding constraints, a concise format and require the model to cite section numbers of the documents used. Finally, the model is asked to answer with a single-sentence fallback for queries that cannot be answered from the retrieved context. These last two elements are particularly important in a regulatory setting, making the answers verifiable against the source document without necessarily knowing exactly which chunks were retrieved and the fallback answer prevents the model from filling gaps with hallucinated content. For the same reason, the temperature is set to a near-deterministic 0.1, reducing the risk of too varied or creative responses.

```
SYSTEM_PROMPT = (
    "You are an expert on Swiss Railway operating regulations in German (FDV - Fahrdienstvorschriften), "
    "French (PCT - Prescriptions de circulation des trains) and Italian (PCT - Prescrizioni sulla circolazione dei
treni).\n"
    "Answer ONLY from the provided excerpts.\n"
    "Be concise: 3-5 sentences is usually enough. Do not repeat the same point.\n"
    "Cite the section numbers and document in your answer.\n"
    "If the provided context is insufficient to answer the question, say so in one sentence."
`
```

Figure 6: Generation system prompt.

Citing the section numbers in the generated answer is possible because each chunk is passed to the model with a **structured header** containing the document title, section number and title, formatted before the chunk text. The **language instruction**, however, is injected at the end of the user message. As there are more than one possible query languages, it could not be specified clearly in the system prompt and the more general instruction *"Answer in the same language as the query."* did not yield consistent results. Especially for Italian queries, even larger models often responded in French. Now the query language is detected automatically using langdetect,[19] which maps the detected ISO code to a language name string appended as a final instruction: *"You must write your entire answer in [Language] only."* The langdetect call includes a fallback for detection failures or unsupported language codes, in which case the instruction reads *"the same language as the question"*, ensuring that the system still produces a usable instruction even when language detection fails.

4. Evaluation Design

This chapter will present the evaluation methodology and expected outcome, using the evaluation data described in chapter 2 above. The main goal is to compare the two RAG setups in a use case of classic Q&A, involving queries in three different languages. A second use case is applied as a stress test of the multilingual pipeline (multi_rag), using cross-lingual terminology queries to probe the appropriateness and limits of this type of setup. In complex multi-component architectures such as RAG pipelines, evaluating only the final output is insufficient: a failure in generation may actually originate in the retrieval stage upstream. Identifying where the pipeline breaks down requires evaluating each component separately.[20]
For both use cases, given the specific context and the lack of previously labelled data, the evaluation is understood to be qualitative rather than quantitative. The evaluation results shall be interpreted as tendencies that might be confirmed by larger studies including more data. On the other hand, the small size of the test set also made human inspection and evaluation feasible, providing a deeper qualitative insight into the generated results.

UC1 - Classic Q&A

The first annotated test set allows us to evaluate the RAG setups separately on three aspects: retrieval performance, terminology precision and overall answer quality.

---

[19] See https://pypi.org/project/langdetect/ for more information.
[20] Huyen 2023, p.200.

First, **retrieval performance** can be measured using the annotated expected target regulation across all languages and for both RAG setups. The queries are processed from the file and the metadata of the two retrieved chunks is compared to each query's expected target regulation and section. The retrieval quality (Precision@K) is thus measured by the number of hits or queries where the retrieved chunks match the expected sections.
The expected outcome here would be that the multilingual setup performs better for queries in French and Italian because the target sections are available in the same language. For German queries, the outcome should be identical for both setups as they are working with the same architecture on the same database. Since lexical search has been integrated into the retrieval pipeline, no negative impact on the retrieval of German queries is expected.
Second, **Generation quality** of the RAGs' answers can be measured in different ways. This project focuses on terminology precision rather than overall answer quality. However, for more complete insight, both aspects have been taken into account here. As we are comparing two RAG setups, this analysis is applied only on queries where retrieval worked both for the mono- and the multilingual setup. Given the expected retrieval failures, writing gold standard answers for all 66 queries would have been disproportionate, reducing the possible evaluation metrics for this aspect.
Terminology precision was measured against the annotated target terminology, i.e. the generated answers were analysed to find out whether the defined target term is present. In order to find out whether the generative model used paraphrasing or different terminology to express the correct concept, the queries were categorised as exact or stem match, which would correspond to a positive hit, semantic match, i.e.paraphrasing or different (wrong) terminology, or no match. Here again, the multilingual setup is expected to perform better for French and Italian queries, because the answer would be based on the same language document featuring the correct terminology. In the monolingual setup, the generative model would be translating a German document into the French or Italian query language without necessarily knowing the correct terminology.

```
JUDGE_SYSTEM = (
        "You are an evaluation assistant. Rate how well the RAG answer responds to the "
        "question based on the reference passage from the source document.\n"
        "Scale: 5 = fully correct and complete and using the correct terminology from "
        "the source document, 4 = mostly correct with minor gaps, 3 = partially correct, "
        "2 = mostly incorrect, 1 = completely wrong or not the same language as the
query.\n"
        "Reply with only: SCORE: <1-5>\nREASON: <one sentence>"
    )
```

Figure 7: LLM-as-judge system prompt.

The **overall answer quality** was measured automatically for all queries using an LLM-as-judge model, applying a simple score between 1 and 5. The model used for this automatic evaluation was *phi4-mini:3.8b*,[21] a small multilingual model available locally through Ollama but with strong reasoning skills. The system prompt includes specific instructions for the scale as well as for the answer format (see Figure 7).
To cross-check the results, the same answers evaluated for terminology precision were also manually evaluated by native language specialists of the SBB Language Service. They were asked to express a preference between the two answers (mono_rag vs. multi_rag) or mark them as equal and justify their evaluation by identifying relevant points such as omissions, hallucinations, incorrect terminology, wrong language or lack of fluency.

UC2 - Cross-lingual terminology queries

For the second use case, since it was intended as an exploratory pipeline stress test, only **terminology precision** was measured, using the second test set. Answers were generated for all queries of the test set and with both RAG setups. The same evaluation was run as for the first test set, i.e. they were matched against the identified target term for each query. The results were again categorised as exact and stem match or no match.
The expected outcome between the two RAG setups is hard to predict, even though the multilingual resources of multi_rag would suggest better results. However, since the queries are completely different from the use case they

[21] See https://ollama.com/library/phi4-mini and https://huggingface.co/microsoft/Phi-4-mini-instruct.

were originally designed for, the question is rather whether they will be able to understand the queries in the first place and then find the necessary resources to formulate a relevant answer.

5. Results and discussion

This chapter will present the results of the evaluation and discuss possible causes or consequences.

UC1 - Retrieval performance

The multilingual RAG setup clearly performed better than the monolingual one when it comes to retrieval, with an overall hit rate of more than 90%, compared to 57.6% for mono_rag. This difference is mainly due to varying performance for different languages: for German both RAGs performed equally well (90.9%), while the hit rate dropped to 50% (it) and even 31.8% (fr) for mono_rag (see Figure 8).

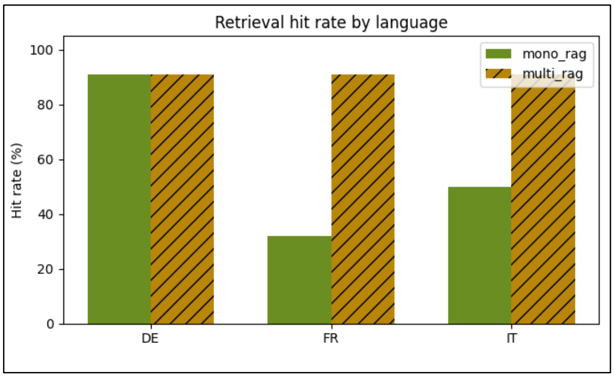

Figure 8: Retrieval performance for all languages.

Only for two queries, both mono_rag and multi_rag were unable to produce the relevant section, even in German. On closer inspection, however, one of them is in fact correct but counted as a miss because of an incorrect section label in the chunk's metadata (section 2.5 instead of 2.5.2). The final results are presented in Table 3.

Language	mono_rag	multi_rag
DE	95.5 %	95.5 %
FR	36 %	95.5 %
IT	54.5 %	95.5 %

Table 3: Retrieval performance for all languages (corrected).

These numbers show the effect of linguistic discrepancies in RAGs: retrieval performance is clearly better when the query and document language match. In other words, when the lexical search cannot be effective because there is no lexical overlap between the languages, semantic search alone will often not be enough to find the correct chunks. As detailed above, this is probably due to the fact that many chunks are semantically very similar. Nevertheless, the fact that for more than half of the Italian queries and a third of the French queries the correct sections are found still shows the impressive baseline of semantic retrieval.

A closer look at the retrieval failures of mono_rag also shows that in some cases, the retrieved sections were completely irrelevant, no obvious semantic link to the query could be determined (see Table 4). It is unclear why these sections were ranked high enough in the retrieval process to even reach the stage of the reranking.

ID	Query	Retrieved section	Generated answer (mono_rag)
26	Comment l'assentiment pour circuler sur les tronçons dépourvus de signaux principaux est-il transmis au mécanicien?	(R 300.9, section 7.2.3) Strassenbahnsignale in Abhängigkeit mit dem Stellwerk Für die Vorbeifahrt an Strassenbahnsignalen, die in Abhängigkeit mit dem Stellwerk stehen, ist bei den Begriffen Ausser Betrieb oder Halt vor dem Befahren eine protokollpflichtige Zustimmung vom FDL erforderlich.	Je suis désolé, mais le texte fourni ne contient aucune information sur la manière dont l'assentiment pour circuler sur les tronçons dépourvus de signaux principaux est transmis au mécanicien. L'extrait 7.2.3 mentionne uniquement la nécessité d'une "protokollpflichtige Zustimmung vom FDL" (approbation documentée du chef de poste) pour la circulation à proximité des aiguilles de tramway dépendantes du poste de contrôle (PCT).
39	Quando è ammesso il trasporto di merci nella cabina di guida occupata?	(R 300.5, section 3.7) usserordentlichen Vorkommnissen gefahren werden, z.B. nach einer Zugtrennung oder im Störungsfall. Das minimal erforderliche Bremsverhältnis muss unter Berücksichtigung der Vorschriften über das Teilbremsverhältnis mindestens für die massgebende Neigung und die gemäss angewendeter Bremstabelle tiefste Geschwindigkeit genügen. Bei Lokzügen richten sich Zug- und Bremsreihe nach der Zulassung der Triebfahrzeuge und dem vorhandenen Bremsverhältnis.	Basandosi esclusivamente sull'estratto fornito, non è possibile rispondere alla domanda. L'estratto non contiene alcuna informazione riguardante il trasporto di merci nella cabina di guida occupata. La sezione 13.4 tratta di situazioni di emergenza e l'uso della frenata di emergenza, mentre la sezione 3.7 riguarda la velocità massima e i rapporti di frenata dei treni.

Table 4: Sample answers based on sections irrelevant to the query.

UC1 - Generation: Terminology precision

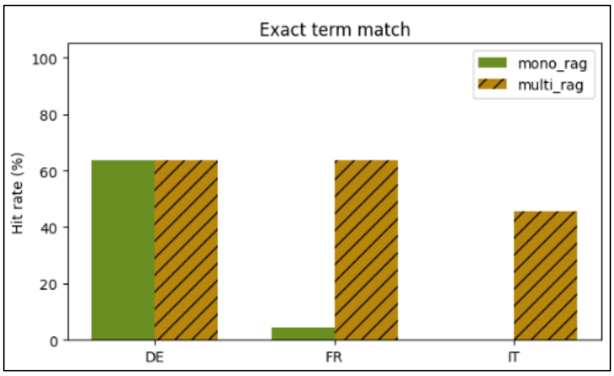

Figure 9: Terminology precision based on exact term matches.

The curated test set also contains target terms allowing us to test whether the generated answers are using the correct terminology. Here, the results are very clear, showing again a direct link between the language mismatch (query and documents) and a failure to produce the correct term in the answer (see Figure 9). While terminology precision is equally high for German, where 63.6% of the answers featured the target term, for both French and Italian answers the mono_rag almost entirely fails to do so.

Again, on closer inspection, the results have to be slightly nuanced, as this first check only involved an exact match against the content of the column "target_term", which would be the correct terminology according to the relevant section. This result therefore does not take into account morphological variation and might explain the lower hit rates. A semantic term match was also added, which would account for any paraphrased or translated terminology as generated directly by the parametric knowledge of the model.

Figure 10 presents the results of this deeper analysis, showing that for French and Italian, mono_rag generates very few answers using the correct terminology, even stem matches included. For both languages, there is some paraphrasing but mostly there is no match, reflecting the high percentage of queries where even the retrieval failed. If we look again only at the answers based on the correct document source (i.e. retrieval hits, see Figure 11), the results are confirmed: only one answer generated by mono_rag in each of those languages features the correct target term (exact and stem match). Given the LLMs' capacity for translation and text generation, more semantic matches would have been expected, either through "free" translation or paraphrasing by the model. It might just be very difficult to recognise a semantic match in this technical domain, depending on how the concept was translated or paraphrased. Interestingly, stem matches seem to be relevant to Italian more than to German, as might have been expected because of the case system or compound nouns.

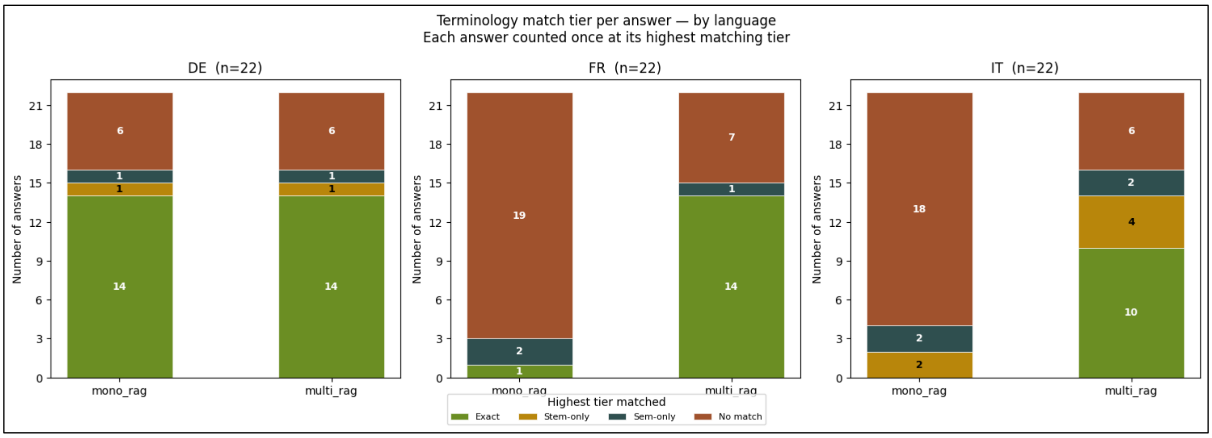

Figure 10: Terminology precision per match type (all queries).

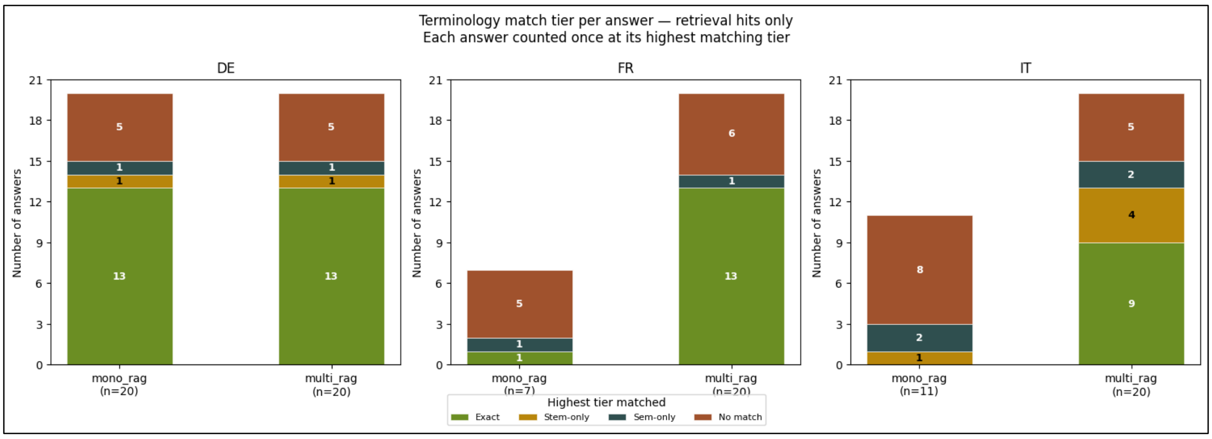

Figure 11: Terminology precision per match type (retrieval hits only).

UC1 - Generation: Answer quality

A first general, manual inspection of the generated answers shows that both RAG setups have been able to create text, meaning that the pipeline is working overall. Only in one case the language of the answer does not match the query language. It concerns an Italian query and the answers generated by both mono_rag and multi_rag are in German. Most likely, the language detection failed because the query is short and contains some very specific vocabulary (*"fuochista"*).[22]

According to the LLM-as-judge, the overall quality of the answers seems quite good, at 3.8 out of 5 for mono_rag and 4.3 for multi_rag. For German, the mean score is 4.5 for both setups while for the other languages, multi_rag performs better than mono_rag (4 vs. 3.8 for French, 4.3 vs. 3.2 for Italian) (see Figure 12).

However, on closer inspection, the automated scores do not always reflect some of the most basic quality requirements: the above-mentioned answer to query ID3 for example was awarded a score of 5 both for mono_rag and multi_rag, despite the language mismatch. A quick check also reveals that there is little consistency in the evaluation of the 28 answers generated by mono_rag without the correct sources (retrieval failure and therefore mostly unusable or unhelpful answers), more than half of them were awarded a score of 4 or 5, while others only

[22] ID 3, interestingly, the answer generated by multi_rag is a backtranslation by the generative model of the Italian document chunk back into German.

1. The mono_rag answer of ID 38 for example, scores 5 with a clearly positive comment,[23] while the answer for ID 39 scores only 1, justified in the comment with a lack of relevance.[24]

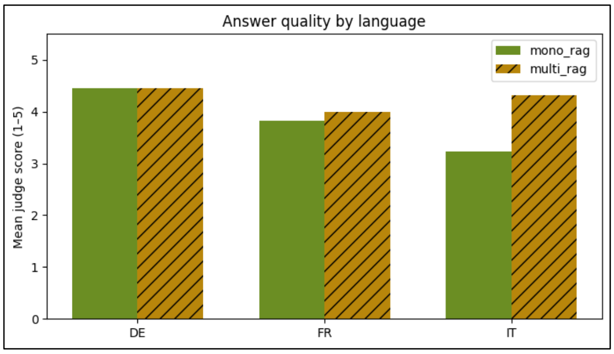

Figure 12: Answer generation quality for all languages.

Even when considering only the answers based on the correctly retrieved sections, the picture remains inconsistent: While for Italian queries the multi_rag answers are still awarded a better mean score (4.2 vs 3.6), and for German the difference remains minimal, for the French queries mono_rag actually scores a lot better than multi_rag (4.6 vs. 3.9, see Figure 13). It thus seems unclear how reliable the automated scores are without in-depth evaluation of the content.

In order to cross-check these results, the generated answers were also evaluated together with the sections provided by the same language specialists involved in reviewing the terminology and queries initially. The results of this evaluation show a different picture: For German, both mono_rag and multi_rag answers are preferred sometimes and in the majority of cases the answers are equal. This is to be expected because both setups have the same documents in German available, sometimes the answers are even identical. For French and Italian, however, there is a clear preference for the answers of the multilingual setup. None of the language specialists has preferred the answer generated by mono_rag for any of the queries (see Figure 14). Among the problems identified are missing references or information from the source segments, especially when lists are involved, additions that are either irrelevant to the query or even completely disconnected (hallucinations) and grammatically wrong or incomplete sentences. For French and Italian, these problems are exacerbated by the presence of incorrect terminology and a mix of languages, especially German words present in the source that are not translated.[25] One language specialist noted that the quality of the generated answers is clearly judged insufficient, some of them containing clear factual errors. Interestingly, the answers generated by multi_rag sometimes seem to start quite well, using the correct terminology, but then the quality and precision deteriorate when the answer gets longer. Also, some of the human evaluations go completely against the LLM's judgement, e.g. ID50, where the answers were awarded scores of 4 (mono_rag) and 1 (multi_rag) while the language specialist clearly preferred the answer generated by multi_rag, judging it to be "correct".

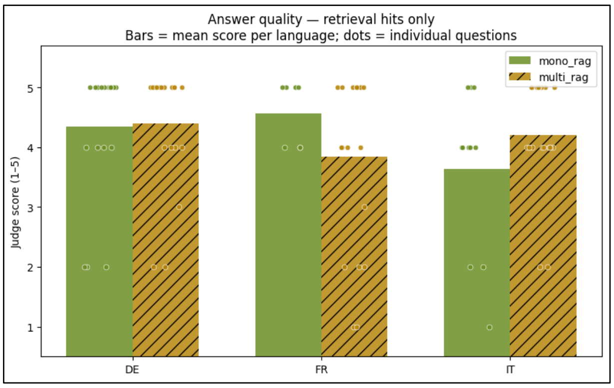

Figure 13: Answer generation quality for retrieval hits only.

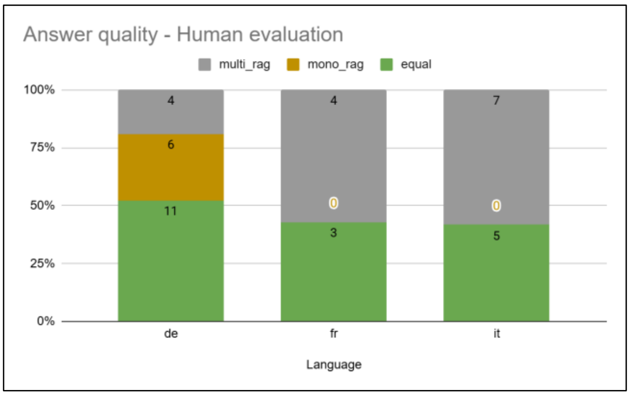

Figure 14: Answer generation quality as evaluated by human language specialists.

The divergence between the automated and human evaluations is itself a finding: the LLM-as-judge consistently failed to penalise answers containing incorrect terminology, language mismatches or

[23] ID 38, mono_rag, score 5: "The answer correctly identifies that the provided text does not contain information about transporting goods in an attendant's cabin, as requested by focusing on sections unrelated to cabins or cargo transportation."
[24] ID 39, mono_rag, score 1: "L'estratto fornito non affronta affatto il trasporto di merci nella cabina di guida occupata; invece, discute solo situazioni di emergenza e le specifiche tecniche della frenata."
[25] For commented examples see annex 3.

unsupported content, awarding high scores to answers that domain experts judged as clearly insufficient. This suggests that automated evaluation metrics might not be best suited to high-stakes specialised domains where terminological precision is a primary quality criterion, and that human expert evaluation remains necessary for meaningful quality assessment in such contexts.
Of course, given the small sample and the constraints imposed by the local setup and hardware capacity, it is difficult to draw general conclusions from these conflicting scores and reviews. It is probable that with larger, more performant LLMs the answer generation but also the automatic judgement would be more consistent and reliable. Nevertheless, it is a reminder that such numbers should not just be taken at face value but verified whenever possible. For these reasons, terminology precision rather than overall answer quality serves as the stronger indicator for the central hypothesis of this project.

UC2 - Cross-lingual evaluation

The evaluation of terminology precision on cross-lingual terminology queries yielded no positive results. The relevant sections could be retrieved for none of the queries and consequently, the terminology check against the target terms also failed for all of the queries. Based on these results, the multi_rag pipeline was adapted incrementally, in order to align it with the different use case:

1. The language filter was moved post-hoc in order to ensure retrieval of target language sections only (i.e. for a query *"Wie heissen Zwergsignale auf Französisch?"* only French language chunks would be retrieved)
2. The BM25 lexical interference was removed (`BM25_WEIGHT=0`), as this would not add any value in cross-lingual queries without lexical overlap.
3. Consequently, the step using the cross-encoder's soft token matching was also removed (`SKIP_RERANK=True`).
4. Finally, the query formulation itself was adapted to a more general question, any references to FDV or PCT were removed (i.e. *"Welcher Begriff wird in den FDV für Sichern verwendet?"* would be adapted to *"Welcher Begriff wird für Sichern verwendet?"*), to avoid retrieving sections mentioning the text.

Despite all these changes, the results were still close to zero: only for one query did the target term feature in the generated answer (multi_rag), but based on the response, the system clearly did not understand the query (see Table 5).

ID	Query	Target term	Generated answer
CL-08	Welche offizielle Abkürzung wird auf Italienisch für den Lokführer verwendet?	MAC	Die bereitgestellten Auszüge enthalten keine Informationen über die offizielle Abkürzung für Lokführer im italienischen PCT. Abschnitt 3.4 beschreibt lediglich die Verwendung von Signalen und die Kommunikation zwischen MAC und CMOV.

Table 5: Sample answer of cross-lingual query.

These results show that the standard RAG recipe (hybrid semantic and lexical search + cross-encoder rerank), as was tuned for same-language factual Q&A queries here, clearly does not hold for this new use case. Crossing the language boundaries, the pipeline clearly breaks down, BM25 getting no overlap and the cross-encoder's attention over shared tokens becoming a liability rather than an asset. Indeed, the embedding model's cross-lingual alignment, while impressive, was never designed to handle translation or terminology questions.
Obviously, a question such as *"Wie heissen Zwergsignale auf Französisch?"* does not aim at retrieving factual content information in the texts but interrogates a linguistic meta-level of the corpus. This changes the entire retrieval paradigm - the answer does not "live" in the chunk but has to be deduced from its structure and therefore requires a completely different approach. This is also the reason why the pipeline fixes have not been enough to yield better results: it is not a problem of tuning the RAG setup, it is a category mismatch between the task and the retrieval architecture itself.

6. Conclusion and Outlook

This project set out to determine whether a multilingual RAG setup produces more precise and consistent terminology than a monolingual one when applied to Swiss railway operating regulations, and to identify the limits of this type of architecture when pushed across language boundaries. The two use cases, classic content Q&A and cross-lingual terminology queries, were designed to test these questions at increasing levels of difficulty.

The retrieval results are the clearest finding of this project. The multilingual setup outperforms the monolingual one decisively for French and Italian queries, with a hit rate of 95.5% compared to 36% and 54.5% respectively for the monolingual setup. This confirms that language matching between queries and documents is a critical factor in retrieval performance, and that semantic search alone is insufficient when many chunks share similar terminology. The hybrid pipeline combining semantic search, BM25 lexical scoring and cross-encoder reranking proved effective for same-language factual queries and is a robust approach for this type of regulatory corpus.

Terminology precision follows directly from retrieval performance. When the correct source document is available in the query language, the generated answers use the correct terminology more consistently. When it is not, the generative model fails to compensate: even with the correct section retrieved, mono_rag almost entirely fails to produce the expected target term for French and Italian queries. This confirms that LLMs cannot be relied upon for domain or regionally specific terminology translation. Their parametric knowledge is simply not precise enough for a context where a wrong term is not just a stylistic imperfection.

The cross-lingual terminology queries produced an interesting and structurally significant finding. Despite incremental adaptations to the pipeline, the system failed completely on all but one query, and even that single result reflected a misunderstanding of the query rather than a genuinely useful answer. The underlying reason is not a tuning problem but a category mismatch: The entire retrieval paradigm assumes the answer lives in a passage, but the questions here do not ask for factual content, they interrogate the linguistic meta-level of the corpus. This is a structural limit of RAG architecture that no amount of pipeline optimisation can resolve and thus necessitates different solutions.

Several conditions of this study limit the generalisability of its conclusions. The generative model used (*gemma3:4b* running locally through Ollama) was selected primarily for practical reasons of hardware capacity and response speed, and its instruction-following and generation quality are clearly below what larger models would achieve today. The test sets, while carefully curated and reviewed by domain experts, are small and the evaluation is qualitative rather than quantitative. The automated LLM-as-judge evaluation also proved unreliable for this domain, often contradicting human judgement. These are important caveats, though they affect the answer quality findings more than the retrieval and terminology precision results, which rest on more objective criteria and remain valid regardless of model size.

Naturally, it would have been informative to run the same evaluation with a larger generative model to clarify how much of the answer quality variance observed here is attributable to model size rather than architecture. Several other extensions are worth pursuing: adding a query translation step to the monolingual pipeline for example could be a low-cost way to improve mono_rag performance, translating French and Italian queries into German before retrieval and running them alongside the original query. Using multilingual data in an aligned form would make cross-lingual retrieval more tractable and could enable parallel lookup.

In the end, every multilingual context is different, and solutions have to be built and evaluated specifically for the situation at hand: the document corpus, the languages, but also the purpose and use cases that apply. For a language service operating in a high-stakes multilingual domain, the findings of this project suggest a clear boundary for what RAG can and cannot usefully do. A multilingual RAG setup with a well-adapted hybrid retrieval pipeline is a viable tool for content Q&A, helping translators understand regulatory content, supporting exam preparation for operational staff, or enabling quick access to specific procedural information across languages. For terminology lookup by language professionals, however, RAG does not seem the appropriate architecture. Dedicated terminology databases integrated in CAT tools remain the better solution for single-term queries. If RAG it must be, for cross-lingual settings where parallel texts are available, a parallel lookup approach (retrieving the relevant chunk and directly accessing its aligned counterpart in the target language) would be both simpler and more reliable than semantic retrieval across language boundaries.

This is also much closer to the approach that standard terminology tools use, but there are other promising approaches emerging. Terminology-Augmented Generation (TAG) for example addresses the language barrier structurally, injecting verified multilingual terminology resources dynamically into the generation process, for example via API access to an existing terminology database. This architecture allows the model to query, filter and apply the correct term directly rather than relying on retrieval to surface it incidentally.[26] The related concept of Lexical-Augmented Generation (LAG) extends this principle beyond terminology to broader lexical and stylistic consistency such as writing guidelines, register or institutional conventions, though the approach is still emerging and not yet fully defined.[27] For an organisation like the SBB that already maintains extensive multilingual terminology resources, in those cases where RAG apparently reaches its limits, TAG might therefore represent a natural and promising next step in the quest for ever more performant tools and resources.

[26] Lackner et al. 2026.

[27] Di Nunzio 2025, p. 102.

Acknowledgements and References

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Throughout this project, Claude Code and Perplexity.ai were used for background research and coding support as well as some language editing of the final report. The conceptual design and methodology of the project as well as the conclusions presented in this report are my own.

Data

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German: https://www.bav.admin.ch/de/fahrdienstvorschriften-fdv
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Annex 1 - Test terminology and data

Complete term list defined to create queries.

Term de	Term fr	Term it	Definition / Interest
Absperrung	barrage	barriera protettiva	Technical, structurally stable device designed to prevent inadvertent intrusion into the danger zone when work is carried out in or near the tracks.
Abstossen, Stoss	lancer, lancée	colpo, colpo	Accelerating vehicles that are being pushed but not coupled with the shunting movement to the required speed and then stopping the shunting movement so that the vehicles continue rolling on their own. The vehicles as they continue to roll are referred to as "Stoss". Very specific terminology to the train operation context: the terms "Stoss", "lancer" and "colpo" would not be considered equivalent in a general language context.
Arbeitsstelle	chantier	area dei lavori	The track area or adjacent location where work is being carried out. The French term "chantier" would be translated as "Baustelle" in German in a general language context. Similar situation with the Italian "area dei lavori" rather than "cantiere".
Einfahrweiche	aiguille d'entrée	scambio d'entrata	First set of points at a station from the direction of line towards the head ("entry turnout" or "entry points"), the corresponding signal is also called "Hirschgeweih" in DE and "signal cactus" in FR. To be distinguished from "Erste Weiche".
Erste Weiche	première aiguille	primo scambio	First set of points of a station that a train will pass when entering, not necessarily an "entry turnout". To be distinguished from "Einfahrweiche".
Heizer	chauffeur	fuochista	Specific function/role on steam engine locomotive. In French the term "chauffeur" is much more general, also used for vehicles like cars, busses or taxis.
Kleinwagen	wagonnet	vagonetto	Towed vehicle without any gear for coupling (buffers, hooks or automatic coupling). Generally smaller than 10t per axle. To be distinguished from "Wagen".
Kreuzung	croisement	incrocio	When two trains pass each other, with one or both continuing on the track used and cleared behind the opposite movement. To be distinguished from "Zugbegegnung".
Lokführer:in (LF)	mécanicien:ne de locomotive (MEC)	macchinista (MAC)	The person responsible for operating traction units of all types with regard to rail operations and technical aspects. Term handled differently according to context or geography: in other French speaking countries (e.g. France) "conducteur de locomotive". At SBB: for recruitment "pilote de locomotive" is also used. In technical documentation it is always "mécanicien de locomotive".
Nachbargleis	voie contiguë	binario adiacente	Adjacent track, the one next to (left or right) from a specific track or work site. To be distinguished from "Nebengleis".
Nebengleis	voie secondaire	binario secondario	A track in a train station with restrictions because of a lack of signals (“secondary” or “side track” in English). To be distinguished from "Nachbargleis".
Neigung: Gefälle und Steigung	déclivité: pente et rampe	pendenza: discesa e salita	Longitudinal profile of a track, terms are different depending on the direction of the train (downward or upward) and can have an impact on the positioning of wagons in the train because they have no brakes.

Perron (das)	quai	marciapiede	Platform next to the rails in a station enabling passengers to board and alight. Specific geographical use: In Switzerland the platform in a train station is called "Perron", the general German term "Bahnsteig" is not permitted. Also, in Switzerland "Perron" takes the neuter gender (das) while according to the Duden, the masculine (der) would also be acceptable.
Sicherheitswärter:in	protecteur:trice	guardiano:a di sicurezza	Different terminology in Italy (it)
Sichern	protéger	assicurare	Taking measures to protect against inadvertent operation those parts of the system that are temporarily unavailable or only to a limited extent available, e.g. to avoid any trains passing on the adjacent track when a train driver has to step out of the train. The measures are taken by the traffic controller at the train driver's request and are not recorded officially. To be distinguished from "Sperren".
Sperren	interdire	sbarrare	Closure of tracks/points for work in the track area. The tracks/points that have been closed cannot be used by trains. Formal procedure, the measures being taken by the safety officer. The procedure (request, confirmation of track closure, reopening the track) is recorded and signed. To be distinguished from "Sichern".
Sperrsignal	signal de barrage	segnale di sbarramento	"Blocking" signal that indicates whether a following section of track may be proceeded on (no exception possible). Intended for trains or other vehicles on tracks. To be distinguished from "Absperrung".
Vorwarner:in	sentinelle	sentinella	Different terminology in Italy (it)
VW (Abk. für Vorwarner:in)	SENT (Sentinelle)	SENT (Sentinella)	In general language documents the abreviation "VW" will probably be used for "Volkswagen" while here it is a specific term (abreviation) for "Vorwarner:in".
Wagen	wagon / voiture	vagone / carrozza	Towed vehicle or carriage with normal gear for coupling or with automatic coupling. To be distinguished from "Kleinwagen". In French and Italian, there is a distinction between carriages for transporting passengers ("voiture" / "carrozza") or goods ("wagon" / "vagone").
Zugbegegnung	rencontre de trains	incontro di treni	When two trains pass each other on adjacent tracks and stay each on their track.
Zwergsignal	signal nain	segnale basso	Ground or dwarf signal. In Italy, the term "marmotta" is used because the signal can be raised out of the ground. NB: the documentation only uses "segnale basso" but in the italian speaking Swiss region of Ticino, the term "segnale nano" is also used, probably in reference to the official German and French terms.

List of queries for Q&A (examples)

ID	Source Term	Query	Language	Relevant section	Target Term
4	Abstossen	Welche Arten von Rangierbewegungen gibt es?	DE	R 300.4, section 1.3	abstossen
5	Abstossen	Quels genres de mouvements de manoeuvre distingue-t-on?	FR	R 300.4, section 1.3	lancer
6	Abstossen	Quali tipi di movimenti di manovra si distinguono?	IT	R 300.4, section 1.3	colpo
7	Perron	Unter welchen Bedingungen sind Rangierfahrten gegen eingestellte Zugfahrstrassen erlaubt?	DE	R 300.4, section 2.3.2	Perron
8	Perron	Dans quelles conditions les courses de manœuvre en direction d'itinéraires de train établis sont-elles autorisées?	FR	R 300.4, section 2.3.2	quai
9	Perron	In quali condizioni sono ammessi i movimenti di manovra verso percorsi treni disposti?	IT	R 300.4, section 2.3.2	marciapiede
10	Kleinwagen	Welche Fahrzeuge dürfen nicht in Züge eingereiht werden?	DE	R 300.5, section 1.4.1	Kleinwagen
11	Kleinwagen	Quels véhicules ne doivent pas être incorporés dans un train?	FR	R 300.5, section 1.4.1	wagonnets
12	Kleinwagen	Quale veicoli non possono essere inseriti in un treno?	IT	R 300.5, section 1.4.1	piccoli veicoli
22	Arbeitsstelle	Unter welchen Bedingungen sind Arbeiten ohne Sicherheitswärter zugelassen?	DE	R 300.12, section 3.1.6	Arbeitsstellen
23	Arbeitsstelle	Dans quelles conditions des travaux sans protecteur sont-ils autorisés?	FR	R 300.12, section 3.1.6	chantiers
24	Arbeitsstelle	Per quali lavori si può rinunciare al guardiano di sicurezza?	IT	R 300.12, section 3.1.6	aree dei lavori
25	Kreuzung	Wie können dem Lokführer die Zustimmung zur Fahrt auf Strecken ohne Hauptsignale erteilt werden?	DE	R 300.6, section 1.3.4	Kreuzung
36	Wagen	Cosa bisogna fare per diminuire i rischi e i pericoli della corrente elettrica?	IT	R 300.8, section 2.4.1	vagone
37	Lokführer	Wann sind Transporte von Waren aller Art im bedienten Führerstand zugelassen?	DE	R 300.13, section 3.2.6	LF
38	Lokführer	Quand est-il permis de transporter des marchandises dans une cabine de conduite desservie?	FR	R 300.13, section 3.2.6	MEC
39	Lokführer	Quando è ammesso il trasporto di merci nella cabina di guida occupata?	IT	R 300.13, section 3.2.6	MAC
40	Gefälle	Wann muss der Lokführer die Wirkung der Luftbremse prüfen?	DE	R 300.14, section 2.3.7	Gefälle
41	Gefälle	À quel moment le mécanicien doit-il s'assurer de l'efficacité du frein à air?	FR	R 300.14, section 2.3.7	pente
42	Gefälle	Quando il macchinista deve verificare l'efficacia del freno ad aria?	IT	R 300.14, section 2.3.7	discesa
43	Neigung	Wie werden Bremskurven ermittelt?	DE	R 300.7, section 3.2	Neigung
44	Neigung	Comment les courbes de freinages sont-elles calculées?	FR	R 300.7, section 3.2	déclivité
45	Neigung	Come sono calcolate le curve di frenatura?	IT	R 300.7, section 3.2	pendenza

ID	Source Term	Query	Language	Relevant section	Target term
46	Zwergsignal	Für welche optischen Signale wird die Farbe weiss verwendet?	DE	R 300.2, section 1.2.1	Zwergsignale
47	Zwergsignal	Pour quels signaux optiques la couleur blanche est-elle utilisée?	FR	R 300.2, section 1.2.1	signaux nains
48	Zwergsignal	Per quali segnali ottici si usa il colore bianco?	IT	R 300.2, section 1.2.1	segnali bassi
49	Erste Weiche	Wo befindet sich die Geschwindigkeitsschwelle im Bahnhof, wenn sich die Bahnhofsgeschwindigkeit auf der Einfahrseite von der Streckengeschwindigkeit unterscheidet?	DE	R 300.6, section 2.2.2	ersten Weiche
50	Erste Weiche	Où se trouve le seuil de vitesse en gare si, du côté de l'entrée, la vitesse en gare est différente de la vitesse de la pleine voie?	FR	R 300.6, section 2.2.2	première aiguille
51	Erste Weiche	Dove si trova la soglia di velocità in stazione se sul lato d'entrata la velocità in stazione è diversa dalla velocità di tratta?	IT	R 300.6, section 2.2.2	primo scambio
52	Nachbargleis	Wann darf im Bahnhof mit einer Höchstgeschwindigkeit von 40 km/h gefahren werden?	DE	R 300.4, section 3.6.3	Nachbargleis e
53	Nachbargleis	Dans quels cas un mouvement de manœuvre peut-il circuler à la vitesse maximale de 40 km/h en gare?	FR	R 300.4, section 3.6.3	voies contiguës
54	Nachbargleis	Quando si può circolare alla velocità massima di 40 km/h in stazione?	IT	R 300.4, section 3.6.3	binari attigui
64	Stoss	Wann ist bei Rangierfahrten das Speichern von Fahrstrassen verboten?	DE	R 300.4, section 2.3.7	Stoss
65	Stoss	Quand est-il interdit d'enregistrer des itinéraires lors des courses de manœuvre?	FR	R 300.4, section 2.3.7	lancer
66	Stoss	Quando è vietata la memorizzazione dei percorsi durante le corse di manovra?	IT	R 300.4, section 2.3.7	colpo

Requests for cross-lingual evaluation (examples).

ID	Source Term	Source Language	Query	Target Language	Gold Term	Relevant Section
CL-01	Abstossen	de	Wie lautet der offizielle französische Fachbegriff für Abstossen?	fr	lancer	R 300.4, section 1.3
CL-02	Lokführer	de	Welche offizielle Abkürzung wird auf Französisch für den Lokführer verwendet?	fr	MEC	R 300.13, section 3.2.6
CL-03	Zwergsignal	de	Wie heissen Zwergsignale auf Französisch?	fr	signaux nains	R 300.2, section 1.2.1
CL-10	Kreuzung	de	Wie lautet der italienische Fachbegriff für Kreuzung?	it	incrocio	R 300.6, section 1.3.4
CL-11	Neigung	de	Wie wird Neigung auf Italienisch bezeichnet?	it	pendenza	R 300.7, section 3.2
CL-12	Sichern	de	Welchen Begriff wird für Sichern verwendet?	it	assicurare	R 300.9, section 4.6.2
CL-15	quai	fr	Comment est désigné en allemand le quai en gare?	de	Perron	R 300.4, section 2.3.2
CL-16	barrage	fr	Quel est le terme allemand pour barrage?	de	Absperrung	R 300.12, section 3.1.7
CL-18	chantiers	fr	Quel est le terme allemand équivalent à chantiers?	de	Arbeitsstellen	R 300.12, section 3.1.6

Annex 2 - Sample results

Queries and generated answers (only retrieval hits) from all languages. Examples are categorised according to comments from the human evaluation.

<table>
<tr><th>ID</th><th>Query</th><th>Target Term</th><th>Answer mono_rag</th><th>Answer multi_rag</th></tr>
<tr><td colspan="5">Incomplete answer, missing references or bullet points.</td></tr>
<tr><td>24</td><td>Per quali lavori si può rinunciare al guardiano di sicurezza?</td><td>aree dei lavori</td><td>Secondo l'estratto 3.1.6, è possibile rinunciare al guardiano di sicurezza (SIWÄ) in determinate circostanze. Nello specifico, si può lavorare senza SIWÄ quando l'intervento prevede al massimo 2 persone, con osservazione continua dei treni e possibilità di evacuazione rapida (ad esempio, addetti alla manutenzione), su tratti con velocità massima di 20 km/h e con sistemi di allarme automatici. Inoltre, è ammesso in aree di lavoro senza necessità di misure di allarme. Tuttavia, è sempre responsabilità del personale di sicurezza pianificare le proprie misure protettive (3.1.6).</td><td>Secondo l'estratto 3.1.6, si può rinunciare al guardiano di sicurezza (GS) esclusivamente per lavori eseguiti da un massimo di due persone, che garantiscano l'osservazione senza ostacoli della circolazione e lo sgombero rapido. Inoltre, è possibile rinunciare al GS sulle sezioni di tratta con corsa a vista e velocità massima di 20 km/h, riconoscendo i pericoli. L'estratto 3.1.5 specifica che si può rinunciare al GS anche per lavori in aree di lavoro senza misure di allarme.</td></tr>
<tr><td>40</td><td>Wann muss der Lokführer die Wirkung der Luftbremse prüfen?</td><td>Gefälle</td><td>Laut Abschnitt 2.3.7 der Fahrdienstvorschriften (Bremsen) muss der Lokführer die Wirkung der Luftbremse unmittelbar nach der Abfahrt und vom Ausgangs- oder Wendebahnhof prüfen (2.3.7). Zusätzlich ist eine Prüfung nach Veränderungen an der Zugzusammensetzung oder nach einem LF-Wechsel sowie vor der Einfahrt in starke Gefälle oder Kopfbahnhöfe erforderlich (2.3.7). Zudem ist eine Bremsprobe nach einer letzten Bremsung mit der automatischen Bremse bei Flugschnee oder großer Kälte vorgeschrieben (2.3.7). Die spezifischen Anforderungen können durch EVU-Betriebsvorschriften ergänzt werden (2.3.7).</td><td>Basierend auf den bereitgestellten Auszügen muss der Lokführer die Wirkung der Luftbremse in folgenden Fällen prüfen:

* Unmittelbar nach der Abfahrt (2.3.7)
* Nach Veränderungen an der Zugzusammensetzung (2.3.7)
* Nach einem LF-Wechsel (2.3.7)
* Vor der Einfahrt in ein starkes Gefälle oder einen Kopfbahnhof (2.3.7)
* Zeitweise nach der letzten Bremsung mit der automatischen Bremse bei Flugschnee oder großer Kälte (2.3.7)</td></tr>
</table>

Hallucinations, completely irrelevant to the query.				
16	Was sind Zielpunkte im Fahrweg?	Einfahrweiche	Laut Abschnitt 4.3.2 der Rangierbewegungen sind Zielpunkte im Fahrweg definiert als Hauptsignale, Gegenverkehrssignale, Sperr- oder Zwergsignale, Anschlussgleise, Zielgleise im Bahnhof oder, bei Bahnhöfen ohne Einfahrsignal, die Einfahrweiche bzw. Bahnhofanfangtafel (4.3.2). Auf zweigleisigen Strecken ohne Wechselbetrieb gilt das Einfahrsignal für beide Gleise als Zielpunkt (4.3.2). Der FDL stellt und sichert den Fahrweg und legt den zugehörigen Zielpunkt fest, abhängig vom gesicherten Fahrweg (4.2.4, 4.3.1). Die Informationen sind ausreichend, um diese Frage zu beantworten.	Laut Abschnitt 4.3.2 der Rangierbewegungen sind Zielpunkte im Fahrweg definiert als Hauptsignale, Gegenverkehrssignale, Sperr- oder Zwergsignale, Anschlussgleise, das Zielgleis im Bahnhof oder, bei Bahnhöfen ohne Einfahrsignal, die Einfahrweiche bzw. Bahnhofanfangtafel (4.3.2). Auf zweigleisigen Strecken ohne Wechselbetrieb gilt das Einfahrsignal für beide Gleise als Zielpunkt (4.3.2). Der Rangierleiter (RL) bestimmt das Zielgleis für jede Rangierbewegung (7.2.4). Vor der Zustimmung zur Rangierbewegung prüft der RL, ob die Weichen korrekt stehen und die Signale die Rangierbewegung erlauben (7.2.5).
36	Cosa bisogna fare per diminuire i rischi e i pericoli della corrente elettrica?	vagone	Secondo l'estratto 2.4.2, per ridurre i rischi e i pericoli derivanti dalla corrente elettrica, è necessario assicurarsi che la cablatura sia spenta, messa a terra e protetta da riavviamenti accidentali prima di qualsiasi attività (PCT 2.4.1). In caso di impossibilità di messa a terra, deve essere coinvolto un SPEA (PCT 2.4.1). È fondamentale evitare il contatto diretto, l'avvicinamento e la manipolazione di connessioni elettriche sotto tensione (PCT 2.4.2). In caso di guasto, il prelievo di corrente deve essere abbassato e la cablatura spenta (PCT 2.4.2).	Basandosi sull'estratto fornito, per diminuire i rischi e i pericoli della corrente elettrica, è necessario assicurarsi che la linea di contatto sia effettivamente disinserita, messa a terra e protetta contro l'inserimento involontario prima di iniziare lavori o salire su veicoli o carichi (Sezione 2.4.1). In caso di impianti elettrici specifici o impossibilità di messa a terra, è obbligatorio rivolgersi a una PESIE (Sezione 2.4.1). È inoltre vietato innaffiare il carbone sui binari con linea di contatto inserita, ad eccezione di specifiche aree consentite (Sezione 3.1). La massima prudenza è raccomandata nel maneggiare attrezzi da fuoco sotto la linea di contatto inserita (Sezione 3.1).

46	Für welche optischen Signale wird die Farbe weiss verwendet?	Zwergsignale	Laut Abschnitt 1.2.1 der Signale (FDV-Signale, Section 1.2: Darstellung) werden Zwergsignale, Weichensignale, Merkzeichen sowie Lichtsignale als Bestätigung oder Ersatz für akustische Signale in der Farbe Weiss dargestellt. Die Farbe Weiss dient somit als allgemeine Kennzeichnung für diese Signalarten. Abschnitt 3.3.1 der Signale (FDV-Signale, Section 3.3.1: Allgemeines) ergänzt dies, indem sie die Verwendung von Weiss für die Bestätigung akustischer Signale durch Licht bestätigt. Die optischen Signale fallen bei Befehlen mit Funk oder mündlich erteilt weg.	Laut Abschnitt 1.2.1 der Signale (FDV-Signale, Section 1.2: Darstellung) werden Zwergsignale, Weichensignale, Merkzeichen sowie Lichtsignale als Bestätigung oder Ersatz für akustische Signale in der Farbe Weiss dargestellt. Die Farbe Weiss dient somit als allgemeine Kennzeichnung für diese Signalarten. Abschnitt 3.3.1 der Signale (FDV-Signale, Section 3.3.1: Allgemeines) ergänzt dies, indem sie die Verwendung von Weiss für die Bestätigung akustischer Signale durch Licht bestätigt. Es ist keine weitere Information in den Exzerpten zur Verfügung gestellt.
60	Cosa si fa prima dell'inizio della prova dei freni?	assicurare	Secondo l'estratto 5.4.3 delle PCT, prima dell'inizio della prova dei freni devono essere completata la formazione del treno e devono essere operativi gli impianto frenante (5.4.3). Inoltre, è necessario che i freni siano rilasciati (5.4.3). Prima della prova, è richiesto che i freni siano rilasciati a bordo di un vagone qualsiasi (4.3.1). Infine, le frenate di sicurezza devono essere rilasciate (4.3.1).	Prima dell'inizio della prova dei freni, secondo le FDV, è necessario che gli apparecchi di frenatura siano pronti ad entrare in funzione (4.3.1). Inoltre, il freno ad aria deve essere allentato su un veicolo qualsiasi, così come i freni di immobilizzazione, ad eccezione di quelli necessari per la sicurezza del treno (4.3.1). Infine, è fondamentale verificare la tenuta stagna degli apparecchi di frenatura (4.3.1).
Grammatically incorrect, incomplete sentences or otherwise strange formulations.				
37	Wann sind Transporte von Waren aller Art im bedienten Führerstand zugelassen?	LF	Transporter von Waren aller Art im bedienten Führerstand sind nur zulässig, wenn sie den Führerstand nicht behindern und der Fluchtweg gewährleistet ist (3.2.6 FDV). Nötig sind die Beförderung der Waren im unbedienten Führerstand, falls dies erforderlich ist (3.2.6 FDV). Es handelt sich um den EVU, die dienstlichen und privaten Warentransporte, die im Führerstand zugelassen sind (3.2.6 FDV). Die Geschwindigkeit darf bei Ausfall der Sicherheitssteuerung nicht höher als 60 km/h betragen (10.4 FDV).	Transporter von Waren aller Art im bedienten Führerstand sind nur zulässig, wenn sie den Führerstand nicht behindern und der Fluchtweg gewährleistet ist (3.2.6 FDV). Nötig sind die Beförderung der Waren im unbedienten Führerstand, falls dies erforderlich ist, um die Bedingungen zu erfüllen (3.2.6 FDV). Es handelt sich um den EVU, also die zugelassenen dienstlichen und privaten Warentransporte im Führerstand (3.2.6 FDV). Die Einhaltung dieser Regelung ist entscheidend für die Sicherheit im Führerstand.

<table>
<tr><td>49</td><td>Wo befindet sich die Geschwindigkeitsschwelle im Bahnhof, wenn sich die Bahnhofsgeschwindigkeit auf der Einfahrseite von der Streckengeschwindigkeit unterscheidet?</td><td>ersten Weiche</td><td>Die Geschwindigkeitsschwelle befindet sich auf der Einfahrseite des Bahnhofs entweder bei der ersten Weiche oder ungefähr 200 Meter nach dem Einfahrsignal, sofern keine Weiche vorhanden ist (2.2.2). Wenn sich im Gleis keine Weiche befindet, beträgt der Abstand 200 Meter (2.2.2). Die Geschwindigkeitsschwelle liegt dann mittig im Perron oder mittig im Bahnhofgebäude, falls kein Perron vorhanden ist (2.2.2). Die Einfahr- und Ausfahrgeschwindigkeit können auf einem weichen- und kurvenfreien Abschnitt gemeinsam befahren werden (2.2.2).</td><td>Die Geschwindigkeitsschwelle befindet sich auf der Einfahrseite des Bahnhofs entweder bei der ersten Weiche oder ungefähr 200 Meter nach dem Einfahrsignal, sofern keine Weiche vorhanden ist (2.2.2). Wenn sich im Gleis keine Weiche befindet, beträgt der Abstand 200 Meter (2.2.2). Die Geschwindigkeitsschwelle liegt dann mittig im Perron oder mittig im Bahnhofgebäude, falls kein Perron vorhanden ist (2.2.2). Die Einfahr- und Ausfahrgeschwindigkeit unterscheiden sich, dann befindet sich die Geschwindigkeitsschwelle mittig im befahrenen Gleis oder mittig im Bahnhof (2.2.2).</td></tr>
<tr><td colspan="5">Wrong terminology-</td></tr>
<tr><td>17</td><td>Quels points peuvent être utilisés comme but du parcours?</td><td>l’aiguille d’entrée</td><td>Selon l'extrait 1, plusieurs points peuvent être utilisés comme point d'arrivée : un aiguillage, un signal principal, un signal de blocage, un aiguillage de raccordement ou le quai de gare (si applicable). De plus, sur une voie biséglée sans trafic mixte, le signal de voie d'approche est également considéré comme un point d'arrivée pour les trajets depuis la voie ferrée vers la gare. L'extrait 2 ne fournit pas d'informations sur les points d'arrivée.</td><td>Selon l’extrait 4.3.2 des Fahrdienstvorschriften, plusieurs éléments peuvent servir de but de parcours : un aiguillage, un signal principal, un signal de barrage, une voie de raccordement ou une voie de but en gare. De plus, sur un tronçon à deux voies sans banalisation, le signal d’entrée sert de but pour les convois circulant vers la gare. L’extrait 4.5.2 précise que l’assentiment doit être transmis en indiquant explicitement le point de départ et le but du parcours.</td></tr>
<tr><td colspan="5">Mix of languages.</td></tr>
<tr><td>21</td><td>In quali condizioni è disponibile lo spazio di sicurezza intermedio?</td><td>binari secondari</td><td>Lo spazio di sicurezza intermedio è disponibile quando esiste un marciapiede, quando è contrassegnato nell'area esterna, tra binari adiacenti o tra un binario e un ostacolo fisso (vedi Sezione 2.5 PCT). Inoltre, è presente se definito nei Regolamenti Operativi dell'ISB o in un dispositivo di sicurezza (vedi Sezione 2.5 PCT). È importante notare che la sua presenza è anche stabilita dalle Betriebsvorschriften (PCT, Sezione 2.5).</td><td>Secondo l'Estratto 1, sezione 2.5, lo spazio di sicurezza intermedio è disponibile in diverse situazioni. È disponibile dove c'è un camminamento, quando è contrassegnato nell'impianto esterno, tra binari secondari o quando è designato nelle prescrizioni d’esercizio del GI. Inoltre, è disponibile quando è menzionato in un dispositivo di sicurezza. L'Estratto 2, sezione 2.1.1, sottolinea che la sua disponibilità è fondamentale per la sosta e i lavori fra i binari.</td></tr>
</table>

47	Pour quels signaux optiques la couleur blanche est-elle utilisée?	signaux nains	Selon l'extrait 1.2.1, la couleur blanche est utilisée pour les zwergsignale, weichensignale, et les merkzeichen, ainsi que comme signal de confirmation ou de remplacement des signaux sonores (1.2.1). Elle est également utilisée pour les signaux optiques en général (1.2.1). Il n'y a aucune information sur l'utilisation de la couleur blanche pour d'autres types de signaux dans cet extrait.	Selon l'extrait 1.2.1, la couleur blanche est utilisée pour les signaux nains, les signaux d'aiguilles et les signaux indicateurs (PCT - Fahrdienstvorschriften, Section 1.2). Elle est également employée pour les signaux lumineux confirmant ou remplaçant des signaux acoustiques (PCT - Fahrdienstvorschriften, Section 1.2). Il n'y a pas d'autres informations concernant l'utilisation de la couleur blanche dans ce document.
48	Per quali segnali ottici si usa il colore bianco?	segnali bassi	Secondo l'estratto 1.2.1, il colore bianco viene utilizzato per i segnali di tipo "Zwergsignale, Weichensignale, Merkzeichen usw." e come conferma o sostituto di segnali acustici (PCT - Signale, Section 1.2: Darstellung). Nell'estratto 2.2, il colore bianco viene impiegato per i segnali che indicano una posizione di "Fahrerlaubnis" (verriegelter Stellung) quando coprono le scheibe o le ruote dei veicoli (PCT - Signale, Section 2.2: Halt-, Kontroll- und Achtungssignale).	Secondo l'Estratto 1, sezione 1.2.1, il colore bianco viene utilizzato per i segnali bassi, gli indicatori e gli altri segnali luminosi che confermano o sostituiscono quelli acustici (1.2.1). L'Estratto 2, sezione 5.1.5, specifica che il segnale principale del sistema N è contrassegnato con un cerchio bianco catarifrangente (5.1.5). Inoltre, i minisegnali principali, utilizzati a velocità inferiori a 40 km/h, si trovano vicino al suolo e possono essere posizionati sopra segnali bassi (5.1.5).
Answer wrong (content).				
23	Dans quelles conditions des travaux sans protecteur sont-ils autorisés?	chantiers	Selon l'extrait 1, les travaux sans protecteur sont autorisés dans des conditions spécifiques. Ils sont permis avec un maximum de 2 personnes, permettant une observation sans restriction des trains et une évacuation rapide et sécurisée. Ces travaux doivent se faire sur des sections de voie où la circulation se fait à vue ou à un maximum de 20 km/h. De plus, ils sont autorisés aux postes de travail équipés d'un système d'alerte automatique et ne nécessitant pas de mesures d'alarme.	Les travaux sans protecteur (PROT) sont autorisés uniquement lorsque le nombre de personnes est limité à deux, assurant une observation complète des convois et une évacuation rapide et sécurisée (par exemple, avec un garde-voie) (3.1.6). Ils doivent également se dérouler sur des tronçons de voie où la circulation se fait en marche à vue à 20 km/h, en raison du concept d'exploitation (3.1.6). Enfin, les personnes effectuant ces travaux sont responsables de leur propre sécurité et doivent planifier leurs propres mesures de protection (3.1.6 & 3.1.6).

Annex 3 - Project structure

The project is available on Github (https://github.com/WpnSta/CAS_Project_tRAG)

Project structure:

Data folder with subfolders for each language (de/fr/it)

Five **notebooks**

- *01_explore.ipynb*: corpus analysis across all three languages.
- *02_mono_rag.ipynb*: complete RAG pipeline indexing only the German documents.
- *03_multi_rag.ipynb*: identical pipeline indexing all three language versions (de/fr/it).
- *04_model_comparison.ipynb*: side-by-side comparison for local LLMs through Ollama.
- *05_QA_evaluation.ipynb*: side-by-side comparison of both setups using test set 1 (UC1).
- *06_crosslingual_eval.ipynb*: exploratory pipeline stress test using test set 2 (UC2).

Python modules with necessary parameters and functions

- *config.py*: parameters shared by all setups and notebooks.
- *chunking.py*: document loading, text cleaning and chunking logic.
- *retrieval.py*: three-pass retrieval pipeline, semantic search via ChromaDB, BM25 hybrid scoring and cross-encoder reranking.
- *generation.py*: language detection, prompt assembly and LLM answer generation via Ollama.

Results folder

- *answer_export.csv*: export of all generated answers for easy inspection (without metadata).
- *crosslingual_results.csv*: export of results from notebook 06.
- *evaluation_results.csv*: export of results from notebook 05.
- *model_comparison_qwen2.5-3b_vs_gemma3-4b.csv*: side-by-side export of answers by 2 modules (notebook 04).